Important links:

facial recognition link:

https://www.kaspersky.com/resource-center/definitions/what-is-facial-recognition

https://www.quora.com/If-someone-used-a-photo-with-my-face-in-front-of-my-laptops-facial-recognition-system-would-they-be-able-to-get-into-my-computer

The iris and retina are both parts of the eye, but they have different function and are not visible in the same way:

Iris

The colored part of the eye that surrounds the pupil. The iris is a muscle that controls the size of the pupil, which in turn controls how much light enters the eye. The iris als contains melanin pigment, which gives the eye its color.

Retina

A thin layer of nerve tissue at the back of the eye that senses light and sends signals the brain. The retina contains many light-sensitive cells that convert light into electric signals that the brain interprets as a visual image. The retina is not visible from the outside of the eye.

The iris and retina can also be used for biometric identification:

Retinal scans

Highly secure, but can be uncomfortable and intrusive because the user needs to position their eye close to the scanner. Retinal scans are often used by high security government agencies.

Iris recognition

Less intrusive than retinal scans and can often be performed at a distance. Iris recognition is more practical for mass scanning scenarios, such as airport security checks.

Retina scans and iris scans are both biometric technologies that use unique patterns in the eye to identify or verify a person:

Retina scans

Use a low-powered infrared light beam to capture the retina's unique patterns. The user must position their eye close to the scanner's eyepiece. Retinal scans are used for security, access control, and financial transactions. However, disease or blindness can affect the accuracy of retinal scans.

Iris scans

Use near-infrared cameras to capture a high-resolution image of the iris from a distance. Iris scans are accurate and precise, with a false acceptance rate of 1:10 million. They can be performed in less than a second and can capture multiple irises at once. Iris scans are used in a variety of settings, including mobile devices, government facilities, and airports. However, some groups may have difficulty using iris recognition technology, such as people with facial disfigurement, blindness, or heavy eye makeup.

The main difference between a coder and a developer is that **a coder focuses on writing code, while a developer is responsible for the entire product development process**:

Coders

Write and test code for programs and mobile apps using programming languages like Java, C#, or Python. Coders are experts at executing specific tasks.

Developers

Responsible for the entire product design process, including researching, prototyping, troubleshooting, documenting, re-engineering, maintenance, and marketing. Developers typically have more experience and training than coders.

Both coders and developers need a strong background in computer science, excellent coding skills, and critical thinking. However, the knowledge required for each role differs, with some overlap.

The maximum distance for iris scanning is currently **up to 40 feet (12 meters) away**. However, iris scanners are typically only effective at a distance of about 1 yard (3 feet).

Iris recognition is a biometric technology that uses a person's iris pattern to identify them. The iris is unique to each person and remains stable over time, making it a reliable method of identification. Iris recognition systems can be used in a variety of applications, including security and law enforcement.

The critical initial recognition step of this process is believed to occur at approximately **50 feet** (Fotios, Uttley, & Fox, 2018). Even at this extended distance, the identity decision is sub-served, in large part, by facial cues rather than body structure (Burton et al., 1999).

Yes, iris scanners can work with glasses, but there are some things to consider:

Strong prescription glasses

Glasses with a strong prescription, sunglasses, or circular contact lenses may prevent the iris scanner from recognizing your irises.

Light reflected from glasses

Light reflected from the surface of your glasses can interfere with the sensor and prevent the scanner from working.

Laser-assisted vision correction surgery

If you've had laser-assisted vision correction surgery, such as LASIK or LASEK, the scanner may not recognize your irises.

Iris recognition is a biometric authentication technology that uses images of a person's iris to identify them. It's considered to be one of the most accurate and flexible biometric authentication technologies. Here are some other features of iris recognition:

- It can work with clear contact lenses and non-mirrored sunglasses.
- It can be used even at night or in the dark because it uses an infrared camera.
- It's contactless, so you don't need to touch the device.
- It's hygienic to use.
- It can distinguish between identical twins.

for who much time

An iris scanner can continuously scan for extended periods, often several hours, depending on its design and power source. However, it's essential to ensure it's well-maintained to avoid any wear and tear. How long are you thinking of using it?

2048-bit SSL encryption is **a strong encryption method that's used to create secure connections between a web browser and a website**. SSL stands for Secure Sockets Layer, a security protocol that encrypts data transmitted over the web. The length of the server's private key determines how difficult it is to crack the encryption.

Here are some things to know about 2048-bit SSL encryption:

- Strength: 2048-bit encryption is strong, but not as strong as a true 2048-bit key.
- Certificates: Most certificate authorities (CAs) only issue 2048-bit certificates.
- Standard SSL: This type of SSL uses SHA-2 and 2048-bit encryption.
- TLS: Transport Layer Security (TLS) is newer and more secure than SSL, but they are functionally the same.

SSL works by:

- Ensuring that the server you're connecting to is the one it claims to be
- Sending a certificate that's vouched for by trusted CAs
- Using session data from the client and server to encrypt and decrypt traffic
- Digitally signing data to verify that it hasn't been tampered with
- Authenticating web servers

Digi locker Link:

https://www.digilocker.gov.in/about/faq

DigiLocker is completely safe and secure to use.

The following practices have been adopted to ensure data security:

Standard Practices: DigiLocker follows standard software development practices of uniform coding standards, guidelines and reviews. Every release is reviewed and tested internally for security and penetration vulnerabilities before getting deployed.

2048 Bit RSA SSL Encryption: DigiLocker uses 2048 bit secure socket layer (SSL) Encryption for information transmitted during any activity.

Multifactor Authentication based Sign Up and Sign in: DigiLocker uses multifactor authentication based signup and signin via Mobile OTP (one time password) or Biometric auth combined with PIN validation for authenticating users and allowing access to the platform.

ISO 27001 certified Data Hosting facilities: The application is hosted on ISO 27001 security certified data hosting facility.

Data Redundancy: Data is backed up in secure environment with proper multi zone redundancy.

Timed Log Out: To protect citizen's account from unauthorized access, our system is designed to terminate session automatically if extended inactivity is detected.

Security Audit: DigiLocker system is audited by CERT-IN empanelled audit agencies and the external third party application security audit certificates are obtained at regular intervals.

User Consent Based System: The data from DigiLocker is shared only with the citizen's explicit consent. All sharing and access activities are logged and conveyed to the citizen. Organizations that need access to citizens' certificates need to register on DigiLocker and seek explicit consent from the citizen.

A pilot project is **a small-scale trial of a project to test its feasibility and concept before a full-scale launch**. Pilot projects can help determine if a project is viable, scalable, and effective, and can provide insights into challenges, costs, and duration.

Pilot projects can be used in many industries, and some examples include:

- Software: Testing a new software release
- Products: Soft launching a new product
- Working hours: Trialing new working hours
- Apps: Testing a new app feature with a small group of users
- AI: Testing a new AI tool with a small subset of customer data

Some benefits of pilot projects include:

Rethinking

If the pilot is unsuccessful, it can be rethought or scrapped without wasting too much time, money, or resources.

Preparation

If the pilot is successful, the wider team or organization can better prepare for big changes.

Refining

The project manager and other stakeholders can refine the approach, strategies, and processes they plan to use for the full-scale implementation.

Difference between trial and pilot

https://www.apm.org.uk/resources/find-a-resource/what-is-the-difference-between-a-trial-and-a-pilot/#:~:text=The%20pilot%20project%20is%20an,is%20new%20to%20the%20organisation.

Definition: Pilot

The pilot project is an initial small-scale implementation that is used to prove the viability of a project idea. This could involve either the exploration of a novel new approach or idea or the application of a standard approach recommended by outside parties but which is new to the organisation. An example of this would be the standard implementation approach for a new off-the-shelf package.

The pilot project enables an organisation to manage the risk of a new idea and identify any deficiencies before substantial resources are committed.

Definition: Trial

The trial forms part of a committed project and is intended to manage the risk of implementation roll out. It tests the implementation approach and its purpose is to manage the risk of implementation.

Firmware link

https://www.fortinet.com/resources/cyberglossary/what-is-firmware#:~:text=Firmware%20Security-,Firmware%20Definition,their%20memory%20to%20function%20smoothly.

Firmware Definition

Firmware is a form of microcode or program embedded into hardware devices to help them operate effectively. Hardware like cameras, mobile phones, network cards, optical drives, printers, routers, scanners, and television remotes rely on firmware built into their memory to function smoothly.

Firmware is often referred to as "software for hardware." However, there is a difference between firmware and software. Firmware provides instructions to help hardware start up, communicate with other devices, and perform basic input/output tasks. Software, on the other hand, is installed onto a device and used for interaction, such as browsing the internet, word processing, listening to music, and videoconferencing.

Hardware manufacturers regularly release firmware updates so their devices remain secure and compatible with new media. For example, Android and iOS devices periodically get firmware updates that enhance their performance, add new features, and protect them against security threats and vulnerabilities. Updates for firmware for devices like game consoles and televisions are typically done by accepting prompts to download and install the latest patch.

Types and Examples of Firmware

When a device switches on, the firmware instructs the processor to initiate the startup process. Computers, mobile phones, and tablets operate multiple forms of firmware to perform actions like loading their operating systems. Firmware operates in a similar way to device drivers. However, firmware is stored directly on the hardware device, whereas drivers are installed in the device's operating system. Firmware can also start up by itself, while drivers remain reliant on the operating system.

There are typically three levels of firmware:

1. **Low-level firmware**: This firmware is usually stored in non-volatile memory chips like read-only memory (ROM) and one-time programmable (OTP) memory. These chips cannot be rewritten or updated, and the firmware is intrinsic to the hardware, such as a computer.
2. **High-level firmware**: This firmware is deployed within flash memory chips and comes with more complex instructions that allow updates to be made.
3. **Subsystems**: These are semi-independent devices that are part of a more extensive system. Firmware at this level is embedded within central processing units (CPUs), flash chips, and liquid crystal display (LCD) units.

Timed Log Out: For Every one minute it scans the iris /face if not matched it get closed or logout.

Important Links: -

1. Microsoft Word - JGI-May-22-ED-005.R3_Toktam_Proof_FINAL.docx (cdspress.ca)
2. (PDF) Internet Gaming Addiction among the Adolescents: A Study from Central India (researchgate.net)

FEASIBILITY AND VIABILITY

- **Market Research and Analysis:** Since our primary audience consists of minors, introducing this software will address a common issue faced by many young people and their parents in the country. By doing so, it aims to enhance their mental well-being and help to shape a better future for them.
- **Potential Challenges:** The primary challenge is the implementation of this software, as it requires access to Voter IDs, necessitating approval from both the Government of India and the Election Commission of India.
- **Strategies to Overcome These Challenges:** To address this, we need to emphasize the importance of the software to government bodies, demonstrating its benefits to gain the necessary approvals for nationwide implementation.

IMPACT AND BENEFITS

Potential Impact on the Target Audience:

- This project can have a significant impact on the target audience.
- In India, approximately 33% of children are addicted to online games, 8.3% of children access unauthorized websites, and reports indicate that children spend up to 6 hours a day on mobile devices.
- Implementing this software aims to address these issues and promote healthier online habits.

Benefits of the Solution:

- The primary benefit of this solution is its positive impact on society.
- By addressing the issue of minors engaging in distracting activities, the software helps ensure that teenagers use their time more productively.
- This not only prevents potential setbacks in their personal development but also mitigates the broader national impact of wasted opportunities and distractions.